

Adaptive Learning Path Recommender System for Cybersecurity Education Using Knowledge Graphs

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Abstract— The increasing complexity of the global cybersecurity landscape, paired with a shortage of 4.8 million qualified professionals, creates an urgency for a paradigm shift in cybersecurity education system. Traditional models such as the one with linear and static learning resources, are increasingly insufficient to address the sheer amount of technological evolution and the vast difference of learner backgrounds. This research proposes the development of an Adaptive Learning Path Recommender System underpinned by a dynamic Knowledge Graph that encompasses the comprehensive cybersecurity domain. Expanding upon existing prior methodologies limited to the cryptography subfield, this study introduces a more refined process for the automated harvesting and structuring of knowledge from a vast array of unstructured data sources, including academic research papers, technical reports, and authoritative internet articles. The methodology leverages keyword extraction and Large Language Models (LLMs) for an accurate entity and relation extraction. Furthermore, the system integrates a hybrid recommender engine that combines knowledge graphs with heuristic path-finding algorithm to generate personalized, goal-oriented learning trajectories. This paper details the entire process from data gathering to graph creation, as well as algorithmic usage required to create this solution, with the end goal of bridging the skill gap through a more dynamic and intelligent system of education.

Keywords—cybersecurity, knowledge graph, education, recommender system

I. INTRODUCTION

The digitization of critical infrastructure, financial systems, and personal data has elevated cybersecurity from a niche technical discipline to a foundational pillar of national security [1] and economic stability[2]. However, this growth of importance is not on par with the supply of workforce indicated by the workforce crisis of unprecedented scale. The 2024 ISC2 Cybersecurity Workforce Studies indicate a global gap of approximately 4.8 million professionals[3], a deficit that continues to widen despite a slight growth in the active workforce. This shortage is more so a qualitative issue than quantitative; organizations are struggling to find personnel with a specific skills required to counter sophisticated threats such as AI-driven attacks [4] advanced malware attacks, and a high effort phishing campaign [5].

The rapid evolution of the cybersecurity field further complicates workforce development. The life expectancy of a knowledge being the latest in the field of cybersecurity is notoriously short due to the number of technologies, vulnerabilities, and attack vectors that are discovered daily. Professionals must possess a "T-shaped" skill set [6], defined as deep expertise in specific areas coupled with a broad

understanding of the threat landscape. Traditional education system which relies on rigid curriculum design processes will struggle to keep pace. By the time a university degree program is updated to include a new technology, the industry standard may have already shifted.

Traditional cybersecurity programs often follow a fixed, linear sequence of courses. While this approach seems logical for beginners, it doesn't account for the learner's current knowledge and can obscure the non-linear connections between concepts. Mastering a topic like web security demands knowledge across multiple domains (programming, networking, system architecture). When courses treat each subject as a linear sequence instead of a non-linear entity, students tend to memorize terms and frameworks without seeing how they fit together [7]. This leads to passive learning, where graduates have theory but struggle to apply skills in novel situations. In other words, a static, document-centric syllabus can create artificial gaps between ideas and leave students unprepared for real threats [8].

Cybersecurity students today come from widely different backgrounds. Veterans of network engineering programs have different prior knowledge than game developers who are switching careers. In practice, one-size-fits-all curricula fail to accommodate this diversity [7]. For example, a network engineer may already understand protocols deeply, whereas a game developer may need that foundation. Training that assumes the same starting point for everyone forces advanced learners to repeat basics and leaves novices behind. Without any kind of personalization, learners can feel lost or waste effort on redundant material. The problem is worsened by the fragmentation of knowledge sources. Cybersecurity learning material is dispersed across multiple platforms: academic databases host theoretical foundations, government repositories (NIST, CISA) provide standards and frameworks, and the "living" knowledge of the field exists in blogs, GitHub repositories, and white papers. Integrating these scattered sources into a coherent learning path is a big cognitive challenge for learners. Previous attempts to map this landscape, such as [9] validated the use of graph-based visualization but were limited by manual curation and narrow scoping, making them difficult to scale to the entire field.

To address these issues in the current academic system, this research proposes a shift towards a data-driven approach to curriculum design. A Knowledge Graph (KG) offers a more dynamic mechanism to model the cybersecurity domain not as a list of topics, but as a network of entities connected by a certain relationship such as the similarity of the articles regarding the topic. This structure can automatically capture the dependencies and prerequisites that are implicit in a field of knowledge, which can be updated periodically to follow

industry trends, as well as assistance for making a curriculum in a low latency manner.

This research expands the methodology of constructing educational knowledge graphs by ingesting data from the entire cybersecurity sources, ranging from high-impact research papers to practical tutorial blogs and extracting the topics and their relationship with each other. The system then constructs a visual representation of these topics that can be dynamically updated as new data comes along. The integration of Large Language Models (LLMs) allows for the extraction of complex relationships with the ability to disambiguate context and link entities with the same meaning

All of this to say, the knowledge graph is a foundation for an Adaptive Learning Path Recommender System. This system can calculate the optimal trajectory for a learner based on their current knowledge state, learning goals, and the difficulty weights of the edges. This facilitates a curriculum that is dynamically generated and adjusted based on the learner's knowledge and industry standard, effectively providing each learner with a personalized syllabus that can prepare them to properly enter the workforce.

II. LITERATURE REVIEW

A. Cybersecurity Standardization Framework

The formalization of cybersecurity education has been driven by the need to define a consistent classification of expertise. Two primary frameworks that are widely used in this landscape are: the National Initiative for Cybersecurity Education (NICE) Workforce Framework [10] and the Cyber Security Body of Knowledge (CyBOK) [11]. The NICE Framework is a comprehensive lexicon for the cybersecurity workforce, mapping specific work roles to specific tasks, knowledge, and skills. While useful for workforce planning and job description alignment, NICE is primarily a functional framework rather than a pedagogical one [12]. It defines what a professional should be able to do, but not the learning required to achieve that competence. [13] highlight the difficulties in mapping university curricula to NICE work roles due to the granularity gap. Academic courses often cover foundational theories for multiple work roles, making a direct mapping challenging.

CyBOK is developed to organize the scientific and engineering foundations of the field, with the approach of dividing the domain into 19 Knowledge Areas (KAs). CyBOK is more academically oriented than NICE, focusing on established principles rather than job tasks. However, both frameworks has an inherently high latency of standardization. They are static documents updated periodically, whereas the threat landscape evolves continuously. A new class of specialty may take a long to be formally incorporated into these standards. This latency creates a curriculum gap where educational institutions teaching to the standard may not cover some of the currently relevant topics.

B. Knowledge Graphs in Educational Technology

The KnowEdu system [14] pioneered the automated construction of educational knowledge graphs by extracting concepts from textbooks and learning assessment data. They utilized neural sequence labeling for entity extraction and association rule mining to identify prerequisite relations, achieving high precision. However, their work focused on K-12 mathematics, a relatively static domain compared to cybersecurity. Within cybersecurity, knowledge graphs have

primarily been deployed for threat intelligence rather than education. Systems like CTIKG integrate data from Indicators of Compromise (IOCs) and multiple CTI articles to help security analysts in performing threat hunting and attribution. [9] attempted to bridge this gap with the "Cryptography Skill Tree," a scoped knowledge graph designed specifically for educational navigation. It demonstrated that clustering keywords from academic papers could create visually intuitive learning maps. However, the methodology was limited by the lack of a dynamic recommender system. This research aims to scale the educational intent using the automated extraction robustness seen in threat intelligence knowledge graphs. [15] propose a methodology to create a knowledge graph for cybersecurity education, but the method would require exponential effort due to the bottom-up approach requiring a human intervention in order to reduce noise and redundant information, as well as contextual disambiguation. With the advancement of LLMs in the past few years, this research aims to leverage such technology to reduce the scaling problem.

C. Keyword Extraction

Named Entity Recognition (NER) method has been rapidly developed, Natural Language Processing (NLP) is often used to extract entities that will potentially be the main keyword for each article. [15] demonstrated using Parts of Speech (POS) tagging and dependency parsing in order to extract entities and their relation. Modern approach for methods to extract technical entities from cybersecurity texts utilizes deep learning, model such as BiLSTM-CRF is used not only to design an educational knowledge graph [16], but also in threat intelligence knowledge graphs [17]. CTINexus however took the approach of experimenting with pretrained transformers such as SecureBERT model and LLMs for direct keyword extraction [18]. LLMs has the advantage of being more context aware, which allows for an automated disambiguation and linking entities.

III. METHODOLOGY

A. Data Collection

The system incorporates many different data sources to ensure a balance between academic literature, industry standardization, and real-time relevance:

- Academic and foundational literature includes peer-reviewed research papers, survey articles, and conference proceedings, these types of sources that are available online are gathered and prepared for data preprocessing phase
- Authoritative frameworks such as CyBOK documentation are also collected and saved in their respective format, this types of documents act as a skeletal structure for the knowledge graph, although the resulting graph might be different from these sources, it will be a useful tool to check if there is any topic that has not been covered by the graph from the lack of sources.
- Community Intelligence such as individual technical blogs, GitHub repositories, and vendor white papers is gathered to capture a more low latency entities in cybersecurity, new technologies, vulnerabilities, or

techniques will appear slower in academic literature compared to online technical blogs.

B. Data Normalization

Harvested data encompasses diverse formats (PDF, HTML, JSON, Markdown) due to the nature of heterogeneous platform source. In order to effectively process the data into a corpus, a preprocessing pipeline is created for major file format, while some file format such as plaintext and Markdown are trivial to parse, other file format requires an additional tool to extract only necessary information, most of which can be dealt with Python libraries, the BeautifulSoup library will be used to remove HTML tags and retain only the main information, the preprocessing of PDF file follows similar route using a different library such as PyPDF.

After converting all the data from multiple file formats into its text form, the tokenization were then applied to the documents for further processing, during this process, stopwords are removed from the text and lemmatization are applied, this minimize the amount of tokens that did not necessarily contribute to the text and improves the computational efficiency.

C. Knowledge Graph Construction

This phase transforms the unstructured corpus into a structured Knowledge Graph, the first step involves extracting the core topics from each collected document. We utilize a domain-specific Named Entity Recognition (NER) model such as CyBERT or KeyBERT to identify technical terms. Once the main topic of each document is decided, the entity linking process will then be performed, which will merge synonyms or abbreviations to each other, this process can be done with LLM, which has the upside of automatically reduce redundant information as well as disambiguating contexts.

Once the labeling process is completed, documents that share the same keyword will then be concatenated and converted into a vector representation with embedding generation algorithms such as TF-IDF, these embeddings can now be compared with one another using algorithms such as cosine similarity, these similarity values will determine how interconnected one topic is with another. In the terms of graph, the similarity value acts as a distance between one node and another. The final result in this process is an interactive graph that can be shown to user from a website where users can zoom and pan across the graph, exploring clusters of related topics.

D. Recommender System

The user can interact with the graph in two ways, the first way is by manually exploring through the graph and some of the nodes that they are familiar with as their current knowledge and defines a destination node as their target interest. With a weighted pathfinding algorithm, it could seek what topics are to learn as a prerequisite to learning their target interest. The second way of using the graph is by interacting with a provided chatbot powered by LLMs that has been tuned with the graph data, using this method, the system can determine the user's current knowledge in a more natural way, as well as getting a better explanation about the syllabus that has been generated based on the knowledge graph.

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